

VANAVARA, Russian Federation

WMO: 249080

Lat: 60.352N

Lon: 102.301E

Elev: 260

StdP: 98.24

Time Zone: 7.00 (E07)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.8	-45.9	-50.9	0.0	-48.7	-48.3	0.0	-45.9	5.1	-19.6	4.4	-17.0	0.3	50	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	29.6	18.0	27.4	17.1	25.3	16.1	19.2	26.7	18.1	24.7	17.2	23.4	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.4	20.7	15.8	11.6	20.3	14.7	10.8	19.5	55.8	26.9	52.5	24.7	49.6	23.3	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
5.0	4.3	3.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-50.5	32.7	3.1	1.7	-52.7	33.9	-54.5	34.9	-56.3	35.9	-58.6	37.1
			-50.3	20.8	3.2	1.2	-52.5	21.7	-54.4	22.3	-56.2	23.0	-58.5	23.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.8	-28.0	-24.4	-12.4	-1.9	6.0	15.3	17.3	13.8	5.8	-3.8	-18.9	-28.1	(6)
(7)		DBStd	17.91	9.74	9.18	8.16	5.88	4.65	4.27	3.32	3.68	4.22	6.48	10.86	11.46	(7)
(8)		HDD10.0	5967	1179	962	693	358	142	7	1	9	138	428	868	1181	(8)
(9)		HDD18.3	8511	1438	1196	952	608	383	108	61	147	376	686	1118	1439	(9)
(10)		CDD10.0	550	0	0	0	0	17	167	226	128	13	0	0	0	(10)
(11)		CDD18.3	52	0	0	0	0	1	18	27	6	0	0	0	0	(11)
(12)		CDH23.3	885	0	0	0	0	11	361	389	121	3	0	0	0	(12)
(13)		CDH26.7	244	0	0	0	0	1	110	102	31	0	0	0	0	(13)
(14)	Wind	WSAvg	1.5	1.1	1.2	1.6	1.9	1.7	1.6	1.5	1.5	1.6	1.4	1.1	(14)	
(15)	Precipitation	PrecAvg	403	20	13	14	25	36	49	53	60	41	34	33	25	(15)
(16)		PrecMax	592	31	41	29	79	64	109	133	153	121	60	66	45	(16)
(17)		PrecMin	272	8	3	1	10	7	8	2	8	4	15	16	10	(17)
(18)		PrecStd	78	6	7	6	14	12	21	32	32	24	12	11	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.1	-1.4	8.2	15.8	25.4	32.2	31.7	30.0	23.0	11.7	3.1	-2.2	(19)
(20)			MCWB	-5.8	-2.6	2.4	6.7	13.7	18.2	19.4	18.8	14.7	7.5	1.7	-2.8	(20)
(21)		2%	DB	-7.4	-3.5	4.7	12.8	21.3	29.8	29.4	26.4	18.7	8.4	1.1	-5.0	(21)
(22)			MCWB	-8.1	-4.6	0.4	5.8	11.4	17.6	18.2	17.0	12.1	5.5	0.2	-5.4	(22)
(23)		5%	DB	-9.8	-6.4	2.1	10.1	18.4	27.6	27.5	23.8	15.7	6.1	-0.4	-7.5	(23)
(24)			MCWB	-10.2	-7.3	-1.0	4.1	9.8	16.7	17.5	15.7	10.9	4.2	-1.3	-7.9	(24)
(25)		10%	DB	-12.8	-9.8	0.0	7.4	15.5	25.2	25.5	21.6	13.1	4.0	-2.5	-10.5	(25)
(26)			MCWB	-13.2	-10.4	-2.4	2.6	8.3	15.4	16.6	14.7	9.8	2.5	-3.2	-10.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.6	-2.5	2.9	7.5	14.6	20.0	20.5	19.7	15.2	8.0	2.0	-2.7	(27)
(28)			MCDWB	-5.2	-1.8	7.1	13.6	23.1	28.9	27.5	27.6	20.5	10.2	2.5	-2.2	(28)
(29)		2%	WB	-8.0	-4.6	0.8	6.0	12.4	18.6	19.5	18.1	13.1	6.0	0.3	-5.4	(29)
(30)			MCDWB	-7.4	-3.5	4.0	12.0	19.2	27.2	26.8	24.1	17.1	8.1	1.0	-5.0	(30)
(31)		5%	WB	-10.3	-7.3	-0.6	4.5	10.7	17.5	18.5	16.9	11.5	4.4	-1.3	-7.9	(31)
(32)			MCDWB	-9.8	-6.3	2.0	9.4	17.1	25.4	24.9	22.3	14.9	5.9	-0.5	-7.5	(32)
(33)		10%	WB	-13.2	-10.5	-2.5	3.0	8.8	16.5	17.7	15.6	10.0	2.7	-3.2	-10.8	(33)
(34)			MCDWB	-12.8	-9.8	-0.1	6.9	15.2	23.4	23.4	20.0	12.9	3.9	-2.4	-10.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	14.5	16.3	12.2	11.4	13.4	12.5	11.8	9.8	7.7	10.2	9.4	(35)
(36)			MCDBR	10.6	10.4	15.1	15.6	17.5	18.5	16.9	16.7	15.3	10.1	6.9	9.0	(36)
(37)			MCWBR	10.4	9.7	11.7	9.4	9.2	7.4	6.9	8.2	9.1	7.4	6.6	9.0	(37)
(38)		5% WB	MCDBR	10.5	10.3	14.0	14.1	15.8	15.9	13.5	13.7	12.5	9.4	7.0	8.8	(38)
(39)			MCWBR	10.3	9.7	11.1	9.0	8.9	6.7	6.1	7.3	7.9	7.1	6.8	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.289	0.315	0.348	0.369	0.396	0.462	0.432	0.341	0.308	0.253	0.224	(40)	
(41)		taud	1.955	2.010	2.047	2.093	2.302	2.339	2.125	2.170	2.433	2.282	2.003	1.900	(41)	
(42)		Ebn at Noon	461	673	792	839	854	834	757	748	782	668	460	336	(42)	
(43)		Edh at Noon	48	84	112	130	115	113	137	121	78	64	45	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.47	1.43	3.00	4.56	5.16	5.73	5.22	4.00	2.40	1.11	0.54	0.26	(44)	
(45)		RadStd	0.03	0.10	0.18	0.30	0.48	0.44	0.41	0.33	0.28	0.12	0.07	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page